CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International General Certificate of Secondary Education

MARK SCHEME for the October/November 2015 series

0620 CHEMISTRY

0620/21 Paper 2 (Core Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2015 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is the registered trademark of Cambridge International Examinations.

Abbreviations used in the Mark Scheme

- ; separates marking points
- / separates alternatives within a marking point
- () the word or phrase in brackets is not required but sets the context
- **A** accept (a less than ideal answer which should be marked correct)
- **I** ignore (mark as if this material were not present)
- **R** reject
- ecf credit a correct statement that follows a previous wrong response
- ora or reverse argument
- owtte or words to that effect (accept other ways of expressing the same idea)

© Cambridge International Examinations 2015

Question	Answer	Marks
1(a)(i)	**F** / barium chloride / $BaCl_2$;	1
1(a)(ii)	**D** / ammonium sulfate / $(NH_4)_2SO_4$;	1
1(a)(iii)	**A** / Al / Al_2Cl_6 / aluminium chloride;	1
1(a)(iv)	**E** / HCl / hydrogen chloride;	1
1(a)(v)	**B** / water / H_2O;	1
1(a)(vi)	**C** / methane / CH_4;	1
1(b)(i)	arrow under the aluminium foil;	1
1(b)(ii)	fume cupboard;	1
1(b)(iii)	3 (Cl_2);	1

Question	Answer	Marks
2(a)	make sure temperature change is the same throughout / make sure that there are no hot spots / no local heating;	1
2(b)	any two from: • same amount of solid / same mass of solid; • same volume of water; • same amount of stirring;	2
2(c)(i)	**Q**	1
2(c)(ii)	**R**; **T**;	2
2(d)(i)	^{235}U;	1
2(d)(ii)	138;	1
2(d)(iii)	cancer treatment / tracer / test thyroid function;	1
2(e)(i)	<u>kerosene</u>;	1
2(e)(ii)	C_5H_{10};	1

© Cambridge International Examinations 2015

Question	Answer	Marks
3(a)	2 (Ni); CO_2 (on right);	1 1
3(b)(i)	positive electrode / anode in box on left; negative electrode / cathode in upper box on right; electrolyte / named suitable electrolyte in lower box on right; 3 correct = [2] 1 or 2 correct = [1]	1 1
3(b)(ii)	cathode / negative electrode;	2
3(c)(i)	*the positive electrode*: chlorine; *the negative electrode*: nickel;	1
3(c)(ii)	inert / unreactive; conducts electricity;	1 1
3(d)(i)	giant structure / lots of carbon atoms joined to each other / lattice of covalent bonds; strong (covalent) bonds throughout;	1 1
3(d)(ii)	weak forces between layers; layers can slide (over each other);	1 1

Question	Answer	Marks
4(a)(i)	gradient / slope is greater for strontium ora;	1

© Cambridge International Examinations 2015

Question	Answer	Marks
4(a)(ii)	11 (cm^3);	1
4(a)(iii)	64–66 (s);	1
4(a)(iv)	the line was still going up / the line was still rising;	1
4(a)(v)	(rate) increases;	1
4(b)(i)	(volumetric) pipette;	1
4(b)(ii)	to show end point of titration / to show when the solution has been neutralised; litmus goes from blue to pink (at end point);	1 1
4(c)(i)	decreases slowly at first; then sudden decrease in pH; then slow decrease;	1 1 1
4(c)(ii)	26 (cm^3);	1
4(c)(iii)	strontium chloride;	1

ROOT 19

© Cambridge International Examinations 2015

Question	Answer	Marks
5(a)	melting / ice melts / ice goes from solid to liquid;	1
	any four from: • in solid particles regularly arranged; • in solid particles arranged in fixed position / cannot move; • particles in solid absorb energy; • particles (in solid) vibrate more / particles start to move when heated; • forces between particles (in solid) broken; • particles in liquid slide over each other / move; • particles in liquid not regularly arranged;	4

ROOT 19

© Cambridge International Examinations 2015

Question	Answer	Marks
5(b)(i)	coolant / for making ethanol / for making specified chemicals / solvent;	1
5(b)(ii)	washing / cooking / cleaning etc.;	1
5(c)(i)	lithium + water → lithium hydroxide + hydrogen;	1
5(c)(ii)	any two from: • floats on surface (of water); • bubbles of gas; • fizzes / fizzing sound; • decreases in size / disappears; • moves around;	2
5(c)(iii)	(potassium) more reactive / lithium less reactive;	1
5(d)(i)	correct structure of ethene;	1
5(d)(ii)	high temperature; catalyst / phosphoric acid;	1 1
5(e)	cobalt → iron → cerium → calcium; one pair reversed or all reversed = [1]	2

Question	Answer	Marks
6(a)	C_5H_8;	1

© Cambridge International Examinations 2015

Question	Answer	Marks
6(b)	bromine / bromine water / aqueous bromine; decolourised / goes from orange to colourless;	1 1
6(c)(i)	double bond;	1
6(c)(ii)	poly(ethene) / any other addition polymer;	1
6(d)	it is a molecule / covalent compound;	1
6(e)	any two from: carbon or soot / carbon monoxide / water;	2
6(f)	alcohols / first box ticked;	1

Question	Answer	Marks
7(a)	any five from: • 11 electrons; • electrons – (negatively) charged; • electrons outside nucleus in shells; • nucleus contains protons and neutrons; • protons – (positively) charged; • neutrons no charge; • 11 protons; • 12 neutrons; • electron arrangement 2,8,1 / 1 electron in outer shell;	5
7(b)(i)	2 (NaOH); 2 (H_2O);	1 1
7(b)(ii)	substance containing carbon and hydrogen only / substance containing carbon and hydrogen and no other element;	1
7(b)(iii)	70;	1

© Cambridge International Examinations 2015